

00 · Cross-Cutting Facts Overview — Master Layer Card (Acceptance Form)

Layer Name

00 · Cross-Cutting (Conventions · Parameters · Input Contract · DoD)

Purpose (Cel)

List all facts that span L1-L10:

- Notation (notation, naming conventions, global numbers 503/510/8 841 820),
- Global numbers in this project (503 tickers/510 zip files/8 841 820 timestamped rows),
- 17 params, input schema, acceptance checklist; single home for each fact.

Input / Output

In: N/A (foundational).

Out: 00_conventions_eng.md, 00_parameters_eng.md, 00_input_contract_eng.md, 00_definition_of_done_eng.md; referenced by every layer card.

See: 00_conventions_eng.md · 00_parameters_eng.md · 00_input_contract_eng.md · 00_definition_of_done_eng.md

Notebook / Artifact

- Canonical params file: config/params.json (exactly 17 keys + EPS + detector constants + L8 thresholds)
- Input contract: OHLCV table schema + price_view + naive-ET→UTC conversion rule (owned in 00_input_contract)
- Definition of Done: 20+ checklist items with layer ownership (owned in 00_definition_of_done)

Tasks / Steps

1. Define notation & naming (00_conventions).
2. Register all 17+ parameters once (00_parameters).
3. Specify table schema + invariants (00_input_contract).
4. Maintain DoD checklist (00_definition_of_done).
5. Every layer references its owning 00_file; never duplicates facts.

QC Gates / DoD Items

- [] Every fact has exactly one home in a 00_file.
- [] On divergence SOT wins.
- [] 503 tickers, 8 841 820 rows, 510 ZIPs referenced consistently.
- [] Causality / determinism / one-shot / gate rules defined once.
- [] All layer cards link back to correct 00_owners.

Dependencies

- Previous: none
- Next: all L1-L10 (reference these)
- Cross-cutting: all 00_* files

3D Visualisation

- Central hub cube or sphere linking all L1-L10 nodes.
- Four walls: conventions, parameters (17 keys), input contract schema, DoD checklist.
- Attached: global numbers panel (503/510/8 841 820) floating above.
- Arrows to each layer node labeled with owning facts.

Footer: Pin this card to 3D model overview node | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/00_*_eng.md | Facts only from SOT.

L1 · Source: Alpaca — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L1 · Source: Alpaca

Purpose (Cel)

Download raw 1h OHLCV for 503 S&P 500 tickers from Alpaca (sip feed) via LEAN CLI; incremental hourly top-ups; store as LEAN ZIPs. Source of truth for all downstream layers.

Input / Output

In: Alpaca Market Data API (REST, sip consolidated tape), config/universe.txt (503 tickers), cron schedule.

Out: 510 ZIP files (<ticker>.zip), one per ticker, CSV rows YYYYMMDD HH:MM,open×10000,high×10000,low×10000,close×10000,volume (naive ET).

See: 00_input_contract_eng.md · L1_source_alpaca_eng.md

Notebook / Artifact

- Tool: QuantConnect LEAN downloader CLI (headless REST client)
- Config files: config/universe.txt + config/params.json (no hardcoded thresholds anywhere)
- Related schedule: hourly cron at :05 ET (session guard: no-op outside RTH)

Tasks / Steps

1. Authorize with Alpaca API key (headless).
2. Download full history for 503 tickers (RTH 09:00-16:00 ET only, ~7 candles/day).
3. Incremental upsert every hour (lookback ~5 days).
4. Guard: outside ET market hours → no-op.
5. Verify volume column present (hard fail QC if absent).

QC Gates / DoD Items

- [] 503 tickers downloaded, 510 ZIPs created.
- [] All CSVs have required columns and ×10000 integer prices.
- [] Timestamps naive ET, no TZ.
- [] Idempotent re-runs produce identical state.
- [] Session guard active (Mon-Fri RTH only).

Dependencies

- Previous: none (L0 = external API, data origin)
- Next: L2 (LEAN ZIP store, first durable archive)
- Cross-cutting owners: 00_conventions_eng.md, 00_parameters_eng.md (global rules and numbers)

3D Visualisation

- Cube node: front wall = Alpaca cloud icon + "L1 Source".
- Left wall: REST API arrow graphic.
- Right wall: 510 ZIP folder icons (one per ticker).
- Back wall: "503 tickers · hourly cron" text panel.
- Attached: incoming data flow lines from external API sphere.

Footer: Pin this card to 3D model layer L1 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L1_source_alpaca_eng.md | Facts only from SOT.

L2 · LEAN ZIP Store — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L2 · LEAN ZIP Store

Purpose (Cel)

Provide durable, compact archival store (510 ZIPs, 139 MB) as the single source of truth for rebuilding the full database; CSV rows without header, prices $\times 10000$, naive ET timestamps.

Input / Output

In: ZIPs from L1 (or rebuild).

Out: 510 <ticker>.zip files; each holds one headerless CSV (YYYYMMDD HH:MM,open $\times 10000$,...); total 8 841 820 rows across all.

See: 00_input_contract_eng.md · L2_lean_zip_store_eng.md

Notebook / Artifact

- Archive format: LEAN zip + CSV row layout
- Size reference: 139 MB total
- Related: only append/replace-whole-ticker edits allowed

Tasks / Steps

1. Receive/validate 510 ZIPs from L1.
2. Enforce name convention <ticker>.zip.
3. Store prices as integers $\times 10000$ (zero FP error).
4. Guard: never partial CSV edit inside a zip.
5. Document: zips are rebuild source for L3 DuckDB.

QC Gates / DoD Items

- [] Exactly 510 ZIPs, 503 universe + extras.
- [] Every CSV row matches YYYYMMDD HH:MM,open,high,low,close,volume (integers $\times 10000$).
- [] Volume column present and integer.
- [] No header row inside CSVs.
- [] Parity with L3 row/symbol counts (P1).

Dependencies

- Previous: L1 (Source, raw download)
- Next: L3 (DuckDB + QC, first analytical store)
- Cross-cutting owners: 00_conventions_eng.md, 00_parameters_eng.md (global rules and numbers)

3D Visualisation

- Cube: front = ZIP archive stack icon.
- Left: L1 arrow + "510 ZIPs".
- Right: CSV rows graphic ($\times 10000$ prices).
- Back: "139 MB · 8 841 820 rows" panel.
- Attached: folder meshes floating around node.

Footer: Pin this card to 3D model layer L2 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L2_lean_zip_store_eng.md | Facts only from SOT.

L3 · DuckDB + VIEW ohlcv_1h + QC-01...QC-11 — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L3 · DuckDB + VIEW ohlcv_1h + QC-01...QC-11

Purpose (Cel)

Canonical analytical database: load zips into raw_ohlcv_1h (integers), expose VIEW ohlcv_1h (USD prices), run QC-01...QC-11 on every load; _meta for provenance.

Input / Output

In: 510 ZIPs from L2.

Out: *.duckdb (166 MB), table raw_ohlcv_1h (8 841 820 rows, 503 symbols), VIEW ohlcv_1h (price /10000), _meta table, QC gate results.

See: 00_input_contract_eng.md · L3_duckdb_raw_view_qc_eng.md

Notebook / Artifact

- DB file: *.duckdb
- View: price_view = raw_usd_view
- QC: 11 predicates (QC-01...QC-11) run on load

Tasks / Steps

1. Load zips → raw_ohlcv_1h (DELETE+INSERT per symbol).
2. Create VIEW ohlcv_1h (USD conversion, zero data copy).
3. Populate _meta (schema_version, built_at, counters).
4. Execute QC-01...QC-11; fail on any violation.
5. Record row/symbol counts for parity (P1).

QC Gates / DoD Items

- [] 8 841 820 rows, 503 symbols, 166 MB DB.
- [] VIEW ohlcv_1h exists and returns USD prices.
- [] All QC-01...QC-11 pass (no NaN/Inf except documented cases).
- [] _meta contains correct provenance.
- [] Uniqueness (symbol, ts) guaranteed.

Dependencies

- Previous: L2 (ZIP store, durable source of truth)
- Next: L4 (Snapshot parquet, read-isolated copy)
- Cross-cutting owners: 00_conventions_eng.md, 00_parameters_eng.md, 00_definition_of_done_eng.md (global rules, numbers, acceptance)

3D Visualisation

- Cylinder DB node or cube: front = DuckDB logo + "VIEW ohlcv_1h".
- Left: ZIP → load arrows.
- Right: USD price view + QC-01...QC-11 table on side wall.
- Back: "8 841 820 rows · 503 symbols · 166 MB" + green checkmarks.
- Attached: small QC predicate list plane.

Footer: Pin this card to 3D model layer L3 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L3_duckdb_raw_view_qc_eng.md | Facts only from SOT.

L4 · Snapshot → Parquet OHLCV — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L4 · Snapshot → Parquet OHLCV

Purpose (Cel)

Atomic read-isolated snapshot of DB materialized as clean per-ticker parquet (zero features); manifest with provenance; enables repeatable downstream transforms.

Input / Output

In: Live DuckDB from L3 (snapshot at point-in-time).

Out: 503 parquet files parquet/<TICKER>/ohlc.parquet (zstd), manifest JSON
(rows/symbols/ts_min/ts_max/price_view), IN-07 flag.

See: L4_snapshot_parquet_eng.md

Notebook / Artifact

- Path: parquet/<TICKER>/ohlc.parquet
- Columns: timestamp · open · high · low · close · volume (USD)
- Manifest: written next to snapshot

Tasks / Steps

1. Take atomic snapshot of DuckDB (retry on torn-read).
2. Write manifest JSON with counters + price_view.
3. COPY ... TO parquet per ticker (zstd compression).
4. Verify 503 files created.
5. Guard: transforms read only snapshot, never live DB.

QC Gates / DoD Items

- [] 503 parquet files, zstd compressed.
- [] Manifest contains IN-07 price_view = raw_usd_view.
- [] Column set exactly matches spec (no extra columns).
- [] Parity with L3 row count.
- [] Snapshot timestamp recorded.

Dependencies

- Previous: L3 (DuckDB, canonical analytical DB)
- Next: L5 (Time split, temporal hygiene)
- Cross-cutting owners: 00_conventions_eng.md (global naming and numbers)

3D Visualisation

- Cube: front = parquet file stack icon.
- Left: DB snapshot arrow.
- Right: 503 ohlc.parquet meshes.
- Back: manifest JSON panel + "zstd".
- Attached: per-ticker folder tree.

Footer: Pin this card to 3D model layer L4 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L4_snapshot_parquet_eng.md | Facts only from SOT.

L5 · Time Split — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L5 · Time Split

Purpose (Cel)

Enforce strict temporal hygiene: three disjoint windows (Warm-up / Train / OOS) per asset with purge (H=24) + embargo (5 sess) buffers; dates from params.json only.

Input / Output

In: Parquet from L4 (Train window for detector).

Out: Windowed series with hard boundaries; purged setup indices; CV scheme (purged walk-forward k=4).

See: L5_time_split_eng.md · 00_parameters_eng.md

Notebook / Artifact

- Splits: config/params.json → splits block
- Warm-up: 2016-01-04 → 2016-10-14 (W=20)
- Train: 2016-10-17 → 2023-12-29
- OOS: 2024-01-02 → 2026-05-29 (frozen)

Tasks / Steps

1. Load per-ticker parquet.
2. Slice into Warm-up / Train / OOS per params.json.
3. Apply PURGE_CANDLES=24 and EMBARGO=5 sessions at boundaries.
4. Verify no label window crosses split (assertion).
5. Emit purged indices for CV.

QC Gates / DoD Items

- [] Splits match params.json exactly (no hardcoded).
- [] Purge/embargo prevent leakage across boundaries.
- [] Warm-up rows with NULL features dropped.
- [] OOS untouched until L10 one-shot.
- [] CV scheme = purged walk-forward (k=4).

Dependencies

- Previous: L4 (Snapshot, atomic read isolation)
- Next: L6 (Detector, setup extraction)
- Cross-cutting owners: 00_parameters_eng.md, 00_definition_of_done_eng.md (split dates and acceptance checklist)

3D Visualisation

- Timeline slab or cube: front = calendar/split icon.
- Left: Warm-up/Train/OOS colored segments.
- Right: purge (H=24) + embargo (5) buffer zones graphic.
- Back: dates from params.json.
- Attached: three window planes with row counts.

Footer: Pin this card to 3D model layer L5 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L5_time_split_eng.md | Facts only from SOT.

L6 · Trend-line Setup Detector (OUTPUT Contract) — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L6 · Trend-line Setup Detector (OUTPUT Contract)

Purpose (Cel)

Bridge raw candles → feature rows: produce setup objects (direction, L_trend, L_opp, t0, ATR, etc.) causally; own the 5 invariants + DET-09; algorithm in companion.

Input / Output

In: parquet/<TICKER>/ohlcv.parquet (Train window, L5).

Out: Setup objects (one per valid break) with 5 invariants; direction (± 1), lines, t0, ATR(t0), R0, entry_candle, etc.

See: L6_setup_detector_eng.md · ../detector_algorithm_eng.md

Notebook / Artifact

- Output contract owned here (5 invariants)
- Reference impl: detector_algorithm_eng.md
- DET-09: reject setups with $R0 \leq 0$ / $ATR(t0) \leq 0$ / missing L_opp

Tasks / Steps

1. Read Train-window parquets.
2. Run detector both directions (long/short).
3. Emit only causal setups (fits $\leq t0$).
4. Enforce 5 invariants on every output object.
5. Count rejected setups (DET-09) in audit.

QC Gates / DoD Items

- [] Every setup satisfies the 5 invariants.
- [] Causality: no future candle used for t0 decision.
- [] DET-09 rejections counted (do not vanish).
- [] Output schema matches contract (table in SOT).
- [] closed_through_line flag correct.

Dependencies

- Previous: L5 (Time split, train window only)
- Next: L7 (Features X + Y, training matrix)
- Cross-cutting owners: 00_definition_of_done_eng.md (causality and detector invariants)

3D Visualisation

- Cube with trend-line graphic on front (resistance/support lines).
- Left: parquet candles → setups.
- Right: setup objects table (direction, L_trend, L_opp, t0, ATR, R0).
- Back: 5 invariants + DET-09 rejection counter.
- Attached: long/short direction arrows.

Footer: Pin this card to 3D model layer L6 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L6_setup_detector_eng.md | Facts only from SOT.

L7 · Features X + Label Y — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L7 · Features X + Label Y

Purpose (Cel)

Transform setups into training matrix: compute 8 causal features at t0, triple-barrier label Y, label_uniqueness_weight; produce Output A/B with exact schema.

Input / Output

In: Setup objects from L6 (Train).

Out: Output A (setups + X + Y), Output B (FEATURE_MANIFEST 7-X + closed_through_line + label); 8 feature formulas, sample weights.

See: L7_features_x_label_y_eng.md · 00_definition_of_done_eng.md

Notebook / Artifact

- 8 features (7 X + closed_through_line audit) at t0 (distance_to_trend_line ... body_to_range_ratio)
- Label Y: first-touch triple-barrier outcome (TP/SL/time, BARRIER_MODE)
- FEATURE_MANIFEST: exactly the 7 X columns + closed_through_line audit flag (Output B)

Tasks / Steps

1. For each setup compute 8 features at t0 (ATR, volume z-score, etc.).
2. Apply triple-barrier labeling (R0 from opposing line).
3. Compute label_uniqueness_weight for overlapping windows.
4. Emit Output A/B with exact column names/types.
5. Guard: closed_through_line = 1 from entry onward.

QC Gates / DoD Items

- [] Output B schema matches L7 spec exactly.
- [] No NaN/Inf beyond documented (volume_z_score=0 when std=0).
- [] label_uniqueness_weight computed correctly.
- [] Y_outcome matches first-touch rule.
- [] Partitioning per {asset_id} × {direction}.

Dependencies

- Previous: L6 (Detector, setup objects)
- Next: L8 (Data Quality Gate, parity & summary)
- Cross-cutting owners: 00_definition_of_done_eng.md (feature manifest and label correctness)

3D Visualisation

- Cube: front = X/Y table icon (8 features + label).
- Left: setups → features.
- Right: floating plane with FEATURE_MANIFEST (7 X cols) + Y + closed_through_line.
- Back: 8 feature formulas or label_uniqueness_weight panel.
- Attached: sample weight matrix near node.

Footer: Pin this card to 3D model layer L7 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L7_features_x_label_y_eng.md | Facts only from SOT.

L8 · Data Quality Gate — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L8 · Data Quality Gate

Purpose (Cel)

Single quality dashboard before training: count parities P1/P2/P3, aggregate L2-L7 stats into summary.json; any FAIL blocks L9; measures only (no fixes).

Input / Output

In: Output B from L7 + prior stores.

Out: summary.json (counters, parities, WARN/FAIL status), L8 dashboard; gate decision (PASS/FAIL).

See: L8_data_quality_eng.md · 00_parameters_eng.md (l8 thresholds)

Notebook / Artifact

- Parity chain: P1 (zip↔DuckDB), P2 (DuckDB↔parquet), P3 (parquet↔Output B)
- summary.json schema owned here
- Thresholds in params.json l8 block

Tasks / Steps

1. Re-count rows/symbols at each store boundary.
2. Assert P1/P2/P3 exact equality (no tolerance).
3. Aggregate counters, compute WARN/FAIL bands.
4. Write summary.json.
5. Gate: FAIL → stop before L9.

QC Gates / DoD Items

- [] P1/P2/P3 all PASS (exact row/symbol match).
- [] summary.json schema correct.
- [] L8 dashboard shows per-layer health.
- [] Any FAIL blocks training.
- [] Reference targets from v1 (8 841 820 rows, 503 symbols).

Dependencies

- Previous: L7 (Features, Output B)
- Next: L9 (Optuna/XGB) — only if PASS (gate blocks training)
- Cross-cutting owners: 00_parameters_eng.md, 00_definition_of_done_eng.md (L8 thresholds and DoD)

3D Visualisation

- Shield/gate cube: front = L8 dashboard icon.
- Left: P1/P2/P3 parity arrows.
- Right: summary.json + counters.
- Back: green PASS / red FAIL wall.
- Attached: 503×metrics preview or parity table.

Footer: Pin this card to 3D model layer L8 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L8_data_quality_eng.md | Facts only from SOT.

L9 · Optuna → XGBoost → strategy.py — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L9 · Optuna → XGBoost → strategy.py

Purpose (Cel)

Tune (200 TPE trials) + train binary:logistic XGB on Train (purged CV); emit self-contained strategy_<TICKER>.py artifact (model base64 + selfcheck); deterministic build.

Input / Output

In: Output B (Train) + non-FAIL L8 gate.

Out: strategy_<TICKER>.py per asset (imports standalone, selfcheck() PASS, hashable), hyperparams from Optuna.

See: L9_optuna_xgboost_eng.md

Notebook / Artifact

- N_TRIALS=200, TPE sampler, MedianPruner
- Objective: AUC-PR, purged walk-forward k=4
- Artifact: base64 model + selfcheck()

Tasks / Steps

1. Guard: OOS never read in this phase.
2. Optuna 200 trials on Train only.
3. Train final XGB with best params + sample weights.
4. Export strategy_*.py (deterministic, hash).
5. Verify selfcheck() and import standalone.

QC Gates / DoD Items

- [] 200 trials completed, no OOS leakage.
- [] selfcheck() PASS on every artifact.
- [] Build deterministic (same input → same hash).
- [] Strategy imports without external deps beyond declared.
- [] L8 was PASS before start.

Dependencies

- Previous: L8 (Gate must PASS, quality dashboard)
- Next: L10 (OOS test, one-shot frozen)
- Cross-cutting owners: 00_definition_of_done_eng.md (deterministic artifact & selfcheck)

3D Visualisation

- Cube: front = Optuna + XGBoost chart icons.
- Left: Output B + non-FAIL gate.
- Right: strategy_<TICKER>.py file meshes (base64 model).
- Back: "200 trials · TPE · AUC-PR" panel.
- Attached: hyperparam table or selfcheck badge.

Footer: Pin this card to 3D model layer L9 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L9_optuna_xgboost_eng.md | Facts only from SOT.

L10 · OOS Test — Master Layer Card (Acceptance Form)

Project: S&P 500 ML Pipeline A (L1-L10) | Print size: A5 landscape | Status: Draft

Layer Name

L10 · OOS Test

Purpose (Cel)

One-shot frozen evaluation on untouched OOS window: run detector + features + model, apply entry rule ($p \geq 0.60$), triple-barrier exits; produce $503 \times$ metrics matrix + distribution report. Never tune after seeing results.

Input / Output

In: Frozen strategy_*.py from L9, OOS parquets (L5).

Out: $503 \times \{\text{PF, Sharpe, MDD\%, TIM\%, WR\%, trades}\}$ matrix (canonical order), hash register, results report.

See: L10_oos_test_eng.md · 00_definition_of_done_eng.md

Notebook / Artifact

- OOS window dates: 2024-01-02 → 2026-05-29 (frozen, never tuned)
- Entry rule: probability $p \geq 0.60$
- Exit rules: TP = R0, SL = moving L_opp(t), time-barrier = 24 candles
- Metrics canonical order in matrix: PF · Sharpe · MDD% · TIM% · WR% (plus trades count)

Tasks / Steps

1. Hash all strategy files → register (freeze).
2. Single OOS run: detector → X features → model → p.
3. Apply entry rule + triple-barrier exits.
4. Compute 503×6 metrics matrix.
5. Produce distribution report; zero tuning after results.

QC Gates / DoD Items

- [] Strategy hashes registered before any OOS run.
- [] OOS result never returns to tuning.
- [] Metrics matrix complete (503 assets).
- [] One-shot only (no iterative OOS peeking).
- [] Artifacts immutable post-hash.

Dependencies

- Previous: L9 (frozen artifacts, hash registered)
- Next: none (end of pipeline, final verdict)
- Cross-cutting owners: 00_definition_of_done_eng.md (OOS one-shot rule)

3D Visualisation

- Cube: front = OOS matrix icon (503 assets).
- Left: frozen strategy.py + OOS window.
- Right: metrics table (PF · Sharpe · MDD · TIM · WR).
- Back: "one-shot · hash registered" panel.
- Attached: results distribution chart plane.

Footer: Pin this card to 3D model layer L10 | See full flow: viz/main_data_flow.html | SOT owner: ENG/Layers_Short_SOT/L10_oos_test_eng.md | Facts only from SOT.